

Legend:

- Chemical signal (Yellow)
- Non-chemical signal (Dark Blue)
- Changes in colony condition (Light Blue)

Event	Preparation	Swarm exodus	Migration	Bivouac	Swarm liftoff	Migration	Colonize new nest
Possible decrease in brood production (7 days)	Yes	No	No	No	No	No	No
Workers engorge (10 days)	Yes	No	No	No	No	No	No
Mass queen rearing (14 days)	Yes	No	No	No	No	No	No
Decrease feeding of queen (14 days)	Yes	No	No	No	No	No	No
Rapid population growth	Yes	No	No	No	No	No	No
Increase vibration of queen (14 days)	No	Yes	No	No	Yes	No	No
Piping of queen (10 days)	No	No	Yes	No	No	No	No
Piping of workers (10 days)	No	No	No	Yes	No	No	No
Waggle dancing for nest sites	No	No	No	No	No	Yes	No
Buzz-running (15 min)	No	No	No	No	No	No	Yes
Waggle dances and stop signals	No	No	No	Yes	No	No	No
Piping of workers & queen (2 hours)	No	No	No	Yes	No	No	No
Vibration of workers (1 hour)	No	No	No	Yes	No	No	No
Buzz-running (40 min)	No	No	No	No	No	No	No

Chemical signals: Possible queen pheromone, Queen pheromone, Queen pheromone, Possible queen pheromone, Queen pheromone, Nasanov pheromone.

Non-chemical signals: Piping of queen, Piping of workers, Waggle dancing for nest sites, Buzz-running (15 min), Streking, Scouts cluster at entrance.

Changes in colony condition: Increase vibration of queen, Piping of queen, Piping of workers, Waggle dancing for nest sites, Buzz-running (15 min), Streking, Scouts cluster at entrance.

Stages: Preparation, Swarm exodus, Migration, Bivouac, Swarm liftoff, Migration, Colonize new nest.

Timeline: Possible decrease in brood production (7 days), Workers engorge (10 days), Mass queen rearing (14 days), Decrease feeding of queen (14 days), Rapid population growth, Increase vibration of queen (14 days), Piping of queen (10 days), Piping of workers (10 days), Waggle dancing for nest sites, Buzz-running (15 min), Waggle dances and stop signals, Piping of workers & queen (2 hours), Vibration of workers (1 hour), Buzz-running (40 min).

Source: Grozinger, Richards, Mattila *Apidologie* 2011

